5

Natural Vegetation and Communities Dependent on Forests

Some of you may be living close to a forest. So you may know about trees, plants, bushes, hills, animals, birds and insects. Share what you know about forests and what activities people do in forests with the class.

Have you ever collected wood, leaves, fruits or flowers from the forest? Share your experience with the class. Make a list of things that people collect from the forest to use in their daily lives. How do they use these things?

We learnt about forests in several chapters in the previous classes. Try and recall what you learnt.

Forests are often mentioned in religious books and folk tales. If you know some of these stories, relate them in class.

Some people believe that forests are the abode of gods and goddesses. Find out about these gods and goddesses and tell the class.

Draw a picture of a forest and compare your drawing with that of the other students.

Forests

People have different notions about forests and attach different values to them. Some are scared of forests. For them, the forests are full of huge trees and wild animals like tigers, bears, snakes, scorpions, etc. But there are others who are not scared of forests. Their children roam around in dense forests without any fear. Some are mesmerized by the beauty of the forests or consider them sacred. Some people see forests as an economic resource that provides raw materials like timber, bamboo, tendu leaves etc – for industries.

People also use forests in different ways. Some people build huts in forests to live in. They hunt small animals and collect fruits for food. Some graze their cattle in forests. For some, forests are tourist spots. Some others want to clear the forests and establish farms, industries and mines.

We must keep in mind that it is not just we human beings who use forests. Plants, trees, grasses, birds, insects etc, too, live and grow in forests. Whenever we think about forests, we must keep in mind the interests of everyone – plant, animal, bird or human beings.

What is a forest? There are many ways to describe and define it.

Write your definition of a forest. Share your definition with others in the class and listen to their definitions. You will see many different definitions emerging.

Make a collective list of the different uses that forests have for human beins and wildlife.

There can be no single definition of a forest. Our definition depends on how we look at forests. For example, we can say that a forest is a large area of land covered by trees and bushes. The question then arises, how big a tract of land? What does being covered by trees mean? How dense? Do plantations of teak, rubber or eucalyptus make a forest? Can there be a forest without animals, birds, insects and humans? All definitions give rise to similar questions.

But we need to move ahead with some understanding of forests. We can say that most forests have the following features:

1. A large tract of land - several kilometres long and wide.
2. Many trees and other vegetation such as grass, bushes, shrubs, creepers etc that grow without human interference.
3. A wide range of bio-diversity, with numerous species existing and propagating naturally without human interference.
4. In India, the forests are home to many tribal communities that have adapted themselves to living in forests without causing damage or destruction.

Do these points define and describe the forests around us?

Are forests important? What will happen if all the forests are cleared to make farms, roads, mines, industries and homes? Can we live without forests? Discuss in class.

5.1 How many types of forests are there? Where are they found?

Where are forests found? This is a difficult question because, thousands of years back, forests covered most of the land on the earth. There were abundant forests wherever there was soil, sunlight and water. Except for the Polar region, the snowcapped mountains, sandy or rocky terrain and deserts with scanty rainfall, forests covered the entire earth. When humans started settling down in villages and towns, they began clearing forests to farm the land. The areas suitable for agriculture, plantations, mines, industries etc were rapidly cleared. By the beginning of the 20thcentury, the only forests remaining were in areas that were not suitable for agriculture, like hill ranges, marshes and rocky terrain. These were regions where people found it difficult to live in. Or else, they were far from human habitations.

Is it right to say the forests abound only in remote hilly and rocky regions?

Identify the forests close to where you stay. Try and find out why these forests were not cleared for farms, mines or cities.

Classification of Forests

Forests can be classified in a number of ways – based on their density, the kind of vegetation they contain, or their legal or administrative status. Based on density, forests can be classified into: very dense forests, dense forests, open forests, degraded forests (scrub), etc.

Administrative classification: The Forest Department classifies forests into three types based on its forest administration system: Reserved forests, protected forests and unclassified forests.

Reserved forests: These are government forests in which no one can cut trees, graze cattle, or collect forest produce even for household use. 54.5% of the forests in our country are reserved forests.

Protected forests: These are managed by the government but it may grant the local people limited right to graze cattle, harvest grass for fodder, and collect fuelwood and minor forest produce in these forests. 29.2% of the forests in our country are protected forests.

Unclassed forests: These are not managed by the forest department and the people have unrestricted right to graze cattle and collect wood etc. in these forests. 16.4% of the forests in our country are unclassed forests.

Forests are also classified on the basis of their vegetation. The kind of trees that grow naturally in any region depends on the temperature, rainfall and climatic conditions of that region. For example, cone bearing (coniferous) trees like pine (*cheed*) and cedar (*deodar*) are found only in very cold climates, where there is heavy snowfall and moisture is available round the year. These trees remain green throughout the year.

If there is moisture throughout the year, but the temperature is warm, then other types of trees grow. These are fruit-bearing, broad-leaved trees whose leaves stay green throughout the year. But in hot climates, where it rains only for a few months in the year, trees such as teak grow. These trees shed their leaves in summer and grow new leaves once it rains.

Therefore, forests can be classified according to the climate of the region and the species of trees that grow in them. On this basis, the forests in India can be classified as follows:

5.1.1. Tropical Evergreen Forests

These forests are found in places where the weather is warm round the year and there is rain almost every month. The annual rainfall is more than 200cm. The trees get moisture and warmth for growth all-round the year. The forests are very dense and abound in many types of trees, plants, creepers, plants that grow on trees (parasites and epiphytes), etc.



Figure 5.1: Evergreen forests of Kerala



Figure 5.2: Sal forests of Bastar - tropical moist forests

They remain green all-round the year, because as soon as a leaf is shed a new leaf replaces it. The forests abound in wildlife. In India, such forests are found in high rainfall areas of the Western Ghats, Lakshadweep, Andaman and Nicobar Islands, and the higher regions of Assam. The vegetation is mostly bamboo, cane, jamun, gurjan, semal, kadam, haldu, seesam, mango etc.

5.1.2. Monsoon Deciduous Forests

Monsoon forests are special to India. Almost 70% of India's forests belong to this category. They are found in areas where the climate is warm and rainfall is limited to a few months. Annual rainfall is in the range of 75cm to 200cm. The trees have broad leaves. In rainfall-deficient months, they shed their leaves to reduce transpiration loss and conserve moisture. New leaves grow once there is enough moisture. Because the trees shed their leaves in the dry months, these forests are also called deciduous forests. Most of the forests in Chhattisgarh fall in this category.

Monsoon forests are divided into two categories based on their rainfall. Where the rainfall is higher, these forests are called moist monsoon forests. Where the rainfall is lower, they are called dry monsoon forests.

Moist Monsoon Forests: These forests are found in areas where rainfall is between 100cm to 200cm. The trees grow very tall. Some trees are massive and evergreen. Shrubs and creepers grow beneath them. These forests are found in most parts of Chhattisgarh, eastern Uttar Pradesh, Bihar, Bengal, Odisha, etc. The predominant tree types are sal, teak, sheesham, amla, neem, mango and chandan, khair, kosum, arjun, etc. Moist monsoon forests are not as dense as the evergreen forests and the trees are also not as tall.

Make a list of the trees you see around you and try to identify whether they are evergreen or deciduous. Remember, all trees shed their leaves, but deciduous trees stay without leaves for several months in the year.

Dry Monsoon Forests: These forests are found in regions where the rainfall is lower - between 70cm to 100cm. Most of the trees shed their leaves in summer. These forests are less dense and have less undergrowth. As a result, grass grows on the land. These forests are common in Punjab, Haryana, East Rajasthan, Gujarat, west and central Madhya Pradesh, Maharashtra and Karnataka. The dominant tree types are teak, tendu, palash, khair, mahua and seesam. Large tracts of forests in these regions have been cleared for agriculture.



Figure 5.3: A Teak Forest during the summer months

5.1.3. Thorn Forests

In regions where the rainfall is less than 70cm the vegetation is mostly thorny shrubs and trees such as babul, ber, khair, etc. The forest is sparse, with grass and thorny shrubs growing in the ample spaces between trees. The space between the trees is taken up by grasses and thorny bushes. Such forests are found in Rajasthan, Haryana, and the desert areas of Gujarat. Only thorny bushes and trees grow in them because of the lack of moisture. The trees have thin trunks and are not so tall.

5.1.4. Mangrove forests

In coastal areas, sea-water constantly flows into the land with the tides. Hence, the soil and water has a high salt content. As a result, such areas have a special type of forest that is called mangrove. The roots of the vegetation in mangroves are specially adapted to deal with the ebb and flow of the tidal seawater. These forests are also called tidal forests. Sundari trees grow in the Ganga and Brahmaputra deltas. The main vegetation of these forests is sundari, kewada and mangroves.

5.1.5. Montane Forests

The vegetation in the mountains is different because of the high altitude and low temperature. At very high altitudes, there is no vegetation and the mountains are covered with snow all the year round. At a little lower, height tender grasses grow when the snow melts in the warmer months. Below this, we mostly have the evergreen, coniferous pine and cedar trees. Still lower, we have mixed forests, where broadleaf trees grow alongside the pines. The oak (banjh) and rohdendraon (buransh) are the most common broadleaf trees in these forests. The rainfall is higher in the eastern parts of the Himalayas compared to the western parts. This is why the Eastern Himalayas have denser forests and greater diversity in vegetation.



Figure 5.4: Himalayan deodar forest

5.2 Forests of Chhattisgarh

23% of India's land mass is covered with forests. This percentage is considered to be less than optimum. The widely accepted minimum is that forests should cover at least one-third of the land. Chhattisgarh has 44% forest cover. Of the state's forests, 43% are reserved forests. People cannot utilize these forests freely . Another 40% are protected forests. People can collect forest produce and graze their cattle in these forests. The remaining 17% are unclassed forests. People can utilize there forest without any restrictions.



Figure 5.5: People who depend on forests for a livelihood



Figure 5.6: People who depend on forests for a livelihool

Chhattisgarh abounds in monsoon deciduous forests because the state has a tropical monsoon climate. Jashpur, Samripat, East Baghelkhand and Dandakaranya are regions with moist monsoon deciduous forests. The Chhattisgarh plains and the western part of the state have dry monsoon deciduous forests. Our state has four types of forests - sal forests, teak forests, mixed forests and bamboo forests.

Sal forests: Bastar is called the island of sal forests. Sal forests are also found in Jaspur, Bilaspur, Kanker and Gariyaband.

Teak forests: The other major forest vegetation in the state is teak. Kawardha (Chilphi Ghati) and Narayanpur (Kursail Ghati) have high quality teak forests. Kanker, Sukma, Dantewada, Dongargarh, and Ambagarh also abound in teak forests.

Mixed forests: Amla, harra, baheda, saja etc are the main vegetation in mixed forests. The Mahanadi basin is home to such forests. They are found in Jashpur, Katghora, Kharsiyan, Mahasamand and the plains of Chhattisgarh.

Bamboo forests: The Sarguja forest division and Kanker forest circle are famous for their Katanga bamboos.

5.3 Utilization and Conservation of Forests

India’s forests have seen such reckless and irresponsible felling of trees over the past 150 years that only 23 percent of the country’s land area is forested today. The forests that remain are also not dense enough. There are many reasons for this.

The two main reasons industrialization and urbanization. Our forests are being over-exploited because of the rising demand for timber, bamboo, etc. One of the responsibilities of the forest department is to control illegal felling of trees by harvesting and selling mature trees in a scientifically planned manner. But the department is always under pressure to increase forest revenues.

In 1980, the forest departments across India earned a surplus of Rs 154,728 crores from the sale of forest produce - after deducting all their operational expenses. The forest department in undivided Madhya Pradesh alone earned a surplus of Rs 49,509 crores from the sale of forest produce.

This clearly shows that forests have been exploited for commercial reasons even in the recent past. The figures quoted are the legal earnings from forests. They do not take into account the large-scale felling of forests by private contractors.

Another important reason for the decreasing forest area is population pressure. In the 60 years of growth and development since India gained independence, we have yet to find alternative livelihoods for the country's expanding population. As a result, the rural people have no option other than agriculture. The only way to find land for farming is by felling trees and clearing forests. The people who feel the pinch the most are the poor. They do not have access to education nor do they have any other resources to earn a livelihood. Those who could not get land for agriculture began grazing livestock in the forests, degrading them further.

The development that has taken place in the country till now has also endangered our forests. Trees were cut on a large scale to lay railway tracks and roads. Large tracts of forest were sacrificed for dams and mines. Thus, forests have paid a high price for the country's development.

We saw at the beginning of the chapter that different people utilize forests in different ways. Among these people are the tribal communities that have been living in forests for generations and have used forest resources without damaging or depleting the forests. Even when they cultivated forest land, they practiced *jhum* cultivation, in which the land was left fallow after a couple of years so it could regenerate into forest. The tribals had a well developed community system to share and utilize forest resources collectively. They sold forest produce for their subsistence, not to earn profits. This saved the forest from over exploitation.

When the British established their rule in India, they began trading in timber on a large scale. Forests were also cleared to bring more land under agriculture and earn more reveue. Earning more revenue is what drove the British decisions regarding forests. This caused extensive damage to the forests. The traditional methods of the tribals to farm forest land and utilize forest produce were dismissed as impractical and uneconomic.

When the British realized how rapidly the forests were depleting, they set up a forest department to take control of all the forests under their rule. The tribals, who earlier owned and used the forest resources collectively now became offenders if the went into the forest to hunt or collect forest produce. The British claimed that it was the uneducated tribals who were the main culprits in depleting the forests and their activities needed to be controlled if the forests were to be protected.

5.4 National Forest Policy and Forest Rights Act 2006

The tribals have been fighting for their forest rights for over 150 years now. This struggle led the Indian Government to frame a National Forest Policy in 1988. This policy acknowledged the crucial role of the tribal and local communities in conservating forests. The policy states that the forests must be utilized in a manner that creates livelihoods for the communities that live near them. It also accepts that the tribals and other forest dependent communities have a right to forest produce for their personal consumption.

A Joint Forest Management Programme was established under this policy. Under this programme, the tribals got the right to collect fodder, fuel wood and minor forest produce and also the right to livelihoods as part of forestry operations. But simultaneously, those farming forest land were pressurized to give up their right to the lands so that the forest could regenerate. These days, large tracts of forests across India are being earmarked for tiger conservation. The tiger conservation activities are supported under Project Tiger and the forests are given the legal status of National Parks. This has led to the displacement of lakhs of tribals.

Worried by the constant impoverization of the tribals, the parliament enacted the Forest Rights Act 2006 after a lot of debate. This act acknowledges, for the first time, that injustice had been done to the tribals for the past 200 years by not recognizing the forest rights they had been enjoying. It also accepts that the forests cannot be conserved and developed without restoring the rights of the tribals.

Under this act, the tribals and other traditional forest-dependent communities were given their traditional forest rights and the owneship of the land they had been cultivating. If this act is properly implemented, the centuries old injustice done to the tribals and other forest dependent communities will be rectified.

While this act was being debated in parliament, many conservationists expressed concern about its possible misuse. They felt that the forest-dependent people may begin exploiting forest produce commercially instead of using it for their personal consumption. This could lead to the rapid depletion of forests. But there were many others who also felt that awarding these rights to the tribals, who had been conserving their forests for centuries, was the only way to protect the forests.

Discuss in class

- **Can the Forests Rights Act 2006 end the injustice to the tribals?**
- **How can the act help in conserving the forests?**
- **What other steps need to be taken?**

Try and understand some of the provisions of the Forest Rights Act 2006 with the help of your teacher. The act gives many rights to communities that traditionally utilize forest resources.

1. Individual and community rights of scheduled tribes and other traditional forest dwellers to homestead land and land for earning a livelihood through agriculture.
2. Right to use forest for fuel, hut building, grazing, food collection etc (Nistar Rights)
3. Ownership rights to minor forest produce and the right to collect and sell such produce by traditional methods.
4. Other community rights to fish and other products from water bodies, as well as livestock grazing rights.
5. Rights of pre-agricultural communities and primitive tribes to housing and house construction.

5.5 Community Initiative for Forest Conservation

How Damodar transformed the forests in his village

Sandh Karmari is a village on the Odisha border located about 45km from Jagdalpur. The forests of sal and other trees are dotted with paddy fields. The Karundi River marks almost the entire state border but in some areas bunds demarcate the border. The settlements of the 8 to 10 communities of the village are fairly widely dispersed.

Even though the paddy fields are extensive, there are forests within the village boundary that are criss-crossed by nallas and rivulets. If you approach the village from any side, you cross paddy fields dotted with mahua trees, which give the landscape a distinctive character. There are also scores of toddy palms that are tapped every morning by toddy vendors.

The monsoon season is also the fishing season, with the nallas abounding with fish. Farm produce such

as chunchunia, chaulai (amaranth) and kochai (colocasia) constitute an important part of the monsoon diet. Boda and mushrooms found under the sal trees are also popular seasonal food. People eat as many as five varieties of jimikand (yam) across the various seasons.

Damodar

Sandh Karmari village was surrounded by barren land 35 years back. The credit for its transformed landscape today goes to Damodar, who was the sarpanch for 35 years until 2009.



Figure 5.7: A ritual puja offered to Mowli Dev

When he returned to the village after completing his studies at Jagdalpur, where he was living in a hostel, Damodar found that all the trees had been cut and people were digging roots to burn as fuel in their homes. Wherever he went, Damodar found only the stumps of trees remaining. "The people had nistar rights to the forests but they sold everything for their subsistence."

Damodar was upset and decided to do something to resolve the problem. He contested for sarpanch in the 1976-77 panchayat elections and won. His first priority as sarpanch was to appoint a few people to protect the forests. People contributed food grain in proportion to the land they owned to create a village fund.



Figure 5.8: The way to the pilgrimage site in Mowli Kot

The degraded nistar forest were regenerated. It is now called 'badla kot' (planted forest). First, grazing in the forest was prohibited. Then, Damodar got seeds from the forest department and planted mango, chironji, mahua, beeja, sal, shatawari, jamun, amla, safed musli, kali musli etc. The 215 acres of forest today stand as proof of what people can do on their own.

Another asset of the people of Sandh Karmari is Mowli Kot - the sacred forest of Mowli Devi. The 100-acre forest is inhabited by langurs, flying squirrels, and numerous birds. It abounds in medicinal plants. We can really appreciate the diversity of an untouched, natural forest in Mowli Kot. At the time, when Damodar became the sarpanch, Mowli Kot was also shrinking. His efforts led to people protecting the sacred forest. They stopped tilling land close to Mowli Kot so that the forest could regenerate and expand. Today, this is the largest forest in Bastar.



Figure 5.9: Regenerated forest of Ghoghre

Green Cover in Rajnandgaon

Ghoghare village is located in the Churia sub-range of the Baghnadi forest range in Rajnandgaon district of Chhattisgarh. The village forest was totally degraded because of timber smuggling, mining and uncontrolled grazing. Today, the forest is lush and green thanks to the sustained efforts of the villagers. The air is clean and the fragrance of flowers is everywhere.

A decade back, the forest was highly degraded. All that was left after illegal felling was a forest of stumps. The villagers couldn't get fodder for their cattle or fuel wood for their homes. Realising the gravity of the situation, they took up the challenge of regenerating the forest. They sought the support of the forest department on the advice of the village elders. A joint forest management committee was formed. Bamboo, medicinal plants and fruit bearing trees were planted.



Figure 5.10. Fodder for Cattle

Today, the once-barren land is populated with a variety of trees. There is a strict community ban on felling trees. The villagers take turns to patrol the forest and protect it from outsiders. They have also appointed a watchman. The patrols report that they see many more wild boar, deer, hyenas etc these days. They now also protect the forest against poachers hunting wildlife.

The forest management committee has planted bamboo in 50 hectares of barren land adjoining the forest. Similarly, the 18–20 households of the Lodhi Bharri basti have planted and protected amla, karanj, teak, bamboo, arjun and mahua in another 50 hectares with adequate soil moisture. So the once barren land now been transformed with green cover. The char, harra, baheda and mahua trees as well as the medicinal plants are now well established and fully grown.

In this way, the hard work and dedication of the residents of forest villages have borne fruit and greenery has returned. The villagers no longer have to spend long hours foraging for fuel and fodder. Everything is available in abundance in the close vicinity of the village itself. The grass grows 4-5ft high in the forest so the cattle have abundant fodder.

Has any such afforestation effort taken place near where you live? If such efforts have taken place, find out the details and discuss them in class.

EXERCISES

1. How many types of forests are there in the administrative classification system?
2. What are the main forest categories in the classification of forests in India?
3. What are the special features of evergreen forests?
4. Where do we find deciduous forests and what are their special features?
5. What are the major tree species found in the forests of Chhattisgarh?
6. What are monsoon forests? Explain in detail?
7. Explain the Indian Forest Policy. List the various measures for forest conservation.
8. What are the main reasons for the degradation of forests in India?

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